

# 6

## Performance Limitations

This chapter introduces some multivariable versions of the Bode's sensitivity integral relations and Poisson integral formula. The sensitivity integral relations are used to study the design limitations imposed by bandwidth constraints and the open-loop unstable poles, while the Poisson integral formula is used to study the design constraints imposed by the non-minimum phase zeros. These results display that the design limitations in multivariable systems are dependent on the directionality properties of the sensitivity function as well as those of the poles and zeros, in addition to the dependence upon pole and zero locations which is known in single-input single-output systems. These integral relations are also used to derive lower bounds on the singular values of the sensitivity function which display the design tradeoffs.

### 6.1 Introduction

One important problem that arises frequently is concerned with the level of performance that can be achieved in feedback design. It has been shown in the previous chapters that the feedback design goals are inherently conflicting, and a tradeoff must be performed among different design objectives. It is also known that the fundamental requirements such as stability and robustness impose inherent limitations upon the feedback properties irrespective of design methods, and the design limitations become more severe in the presence of right-half plane zeros and poles in the open-loop transfer function.

An important tool that can be used to quantify feedback design constraints is furnished by the Bode's sensitivity integral relation and the Poisson integral formula. These integral formulae express design constraints directly in terms of the system's sensitivity

and complementary sensitivity functions. A well-known theorem due to Bode states that for single-input single-output open-loop stable systems with more than one pole-zero excess, the integral of the logarithmic magnitude of the sensitivity function over all frequencies must equal to zero. This integral relation therefore suggests that in the presence of bandwidth constraints, the desirable property of sensitivity reduction in one frequency range must be traded off against the undesirable property of sensitivity increase at other frequencies. A result by Freudenberg and Looze [1985] further extends Bode's theorem to open-loop unstable systems, which shows that the presence of open-loop unstable poles makes the sensitivity tradeoff a more difficult task. In the same reference, the limitations imposed by the open loop non-minimum phase zeros upon the feedback properties were also quantified using the Poisson integral. The results presented here are some multivariable extensions of the above mentioned integral relations.

We shall now consider a linear time-invariant feedback system with an  $n \times n$  loop transfer matrix  $L$ . Let  $S(s)$  and  $T(s)$  be the sensitivity function and the complementary sensitivity function, respectively

$$S(s) = (I + L(s))^{-1}, \quad T(s) = L(s)(I + L(s))^{-1}. \quad (6.1)$$

Before presenting the multivariable integral relations, recall that a point  $z \in \mathbb{C}$  is a transmission zero of  $L(s)$ , which has full normal rank and a minimal state space realization  $(A, B, C, D)$ , if there exist vectors  $\zeta$  and  $\eta$  such that the relation

$$\begin{bmatrix} A - zI & B \\ C & D \end{bmatrix} \begin{bmatrix} \zeta \\ \eta \end{bmatrix} = 0$$

holds, where  $\eta^* \eta = 1$ , and  $\eta$  is called the *input zero direction* associated with  $z$ . Analogously, a transmission zero  $z$  of  $L(s)$  satisfies the relation

$$\begin{bmatrix} x^* & w^* \end{bmatrix} \begin{bmatrix} A - zI & B \\ C & D \end{bmatrix} = 0,$$

where  $x$  and  $w$  are some vectors with  $w$  satisfying the condition  $w^* w = 1$ . The vector  $w$  is called the *output zero direction* associated with  $z$ . Note also that  $p \in \mathbb{C}$  is a pole of  $L(s)$  if and only if it is a zero of  $L^{-1}(s)$ . By a slight abuse of terminology, we shall call the input and output zero directions of  $L^{-1}(s)$  the input and output pole directions of  $L(s)$ , respectively.

In the sequel we shall preclude the possibility that  $z$  is both a zero and pole of  $L(s)$ . Then, by Lemma 3.27 and 3.28,  $z$  is a zero of  $L(s)$  if and only if  $L(z)\eta = 0$  for some vector  $\eta$ ,  $\eta^* \eta = 1$ , or  $w^* L(z) = 0$  for some vector  $w$ ,  $w^* w = 1$ . Similarly,  $p$  is a pole of  $L(s)$  if and only if  $L^{-1}(p)\eta = 0$  for some vector  $\eta$ ,  $\eta^* \eta = 1$ , or  $w^* L^{-1}(p) = 0$  for some vector  $w$ ,  $w^* w = 1$ .

It is well-known that a non-minimum phase transfer function admits a factorization that consists of a minimum phase part and an all-pass factor. Let  $z_i \in \mathbb{C}_+$ ,  $i = 1, \dots, k$ , be the non-minimum phase zeros of  $L(s)$  and let  $\eta_i, \eta_i^* \eta_i = 1$ , be the input directions generated from the following iterative procedure

- Let  $(A, B, C, D)$  be a minimal realization of  $L(s)$  and  $B^{(0)} := B$ ;
- Repeat for  $i = 1$  to  $k$

$$\begin{bmatrix} A - z_i I & B^{(i-1)} \\ C & D \end{bmatrix} \begin{bmatrix} \zeta_i \\ \eta_i \end{bmatrix} = 0$$

$$B^{(i)} := B^{(i-1)} - 2(\operatorname{Re} z_i) \zeta_i \eta_i^*.$$

Then, the *input factorization* of  $L(s)$  is given by

$$L(s) = L_m(s) B_k(s) \cdots B_1(s)$$

where  $L_m(s)$  denotes the minimum phase factor of  $L(s)$ , and  $B_i(s)$  corresponds to the all-pass factor associated with  $z_i$ :

$$B_i(s) = I - \frac{2\operatorname{Re} z_i}{s + \bar{z}_i} \eta_i \eta_i^*. \quad (6.2)$$

In fact,  $L_m(s)$  can be written as

$$L_m(s) := \left[ \frac{A}{C} \middle| \frac{B^{(k)}}{D} \right].$$

For example, suppose  $z \in \mathbb{C}$  is a zero of  $L(s)$ . Then it is easy to verify using state space calculation that  $L(s)$  can be factorized as

$$L(s) = \left[ \frac{A}{C} \middle| \frac{B - 2(\operatorname{Re} z) \zeta \eta^*}{D} \right] \left( I - \frac{2\operatorname{Re} z}{s + \bar{z}} \eta \eta^* \right).$$

Note that a non-minimum phase transfer function admits an *output factorization* analogous to the input factorization, and the subsequent results can be applied to both types of factorizations.

## 6.2 Integral Relations

In this section we provide extensions of the Bode's sensitivity integral relations and Poisson integral relations to multivariable systems. Consider a unit feedback system with a loop transfer function  $L$ . The following assumptions are needed:

**Assumption 6.1** *The closed-loop system is stable.*

**Assumption 6.2**

$$\lim_{R \rightarrow \infty} \sup_{\substack{s \in \bar{\mathbb{C}}_+ \\ |s| \geq R}} R \bar{\sigma}(L(s)) = 0$$

Each of these assumptions has important implications. Assumption 6.1 implies that the sensitivity function  $S(s)$  is analytic in  $\overline{\mathbb{C}}_+$ . Assumption 6.2 states that the open-loop transfer function has a rolloff rate of more than one pole-zero excess. Note that most of practical systems require a rolloff rate of more than one pole-zero excess in order to maintain a sufficient stability margin. One instance for this assumption to hold is that each element of  $L$  has a rolloff rate of more than one pole-zero excess.

Suppose that the open-loop transfer function  $L(s)$  has poles  $p_i$  in the open right-half plane with input pole directions  $\eta_i$ ,  $i = 1, \dots, k$ , which are obtained through a similar iterative procedure as in the last section.

**Lemma 6.1** *Let  $L(s)$  and  $S(s)$  be defined by (6.1). Then  $p \in \mathbb{C}$  is a zero of  $S(s)$  with zero direction  $\eta$  if and only if it is a pole of  $L(s)$  with pole direction  $\eta$ .*

**Proof.** Let  $p$  be a pole of  $L(s)$ . Then there exists a vector  $\eta$  such that  $\eta^* \eta = 1$  and  $L^{-1}(p)\eta = 0$ . However,  $S(s) = (I + L(s))^{-1} = (I + L^{-1}(s))^{-1} L^{-1}(s)$ . Hence,  $S(p)\eta = 0$ . This establishes the sufficiency part. The proof for necessity follows by reversing the above procedure.  $\square$

Then the sensitivity function  $S(s)$  can be factorized as

$$S(s) = S_m(s)B_1(s)B_2(s) \cdots B_k(s) \quad (6.3)$$

where  $S_m(s)$  has no zeros in  $\overline{\mathbb{C}}_+$ , and  $B_i(s)$  is given by

$$B_i(s) = I - \frac{2\text{Rep}_i}{s + \overline{p}_i} \eta_i \eta_i^*.$$

**Theorem 6.2** *Let  $S(s)$  be factorized in (6.3) and suppose that Assumptions 6.1-6.2 hold. Then*

$$\int_0^\infty \ln \overline{\sigma}(S(j\omega)) d\omega \geq \pi \lambda_{\max} \left( \sum_{j=1}^k (\text{Rep}_j) \eta_j \eta_j^* \right). \quad (6.4)$$

It is also instructive to examine the following two extreme cases.

(i) If  $\eta_i^* \eta_j = 0$  for all  $i, j = 1, \dots, k$ ,  $i \neq j$ , then

$$\int_0^\infty \ln \overline{\sigma}(S(j\omega)) d\omega \geq \pi \max \text{Rep}_i. \quad (6.5)$$

(ii) If  $\eta_i = \eta$  for all  $i = 1, \dots, k$ , then

$$\int_0^\infty \ln \overline{\sigma}(S(j\omega)) d\omega \geq \pi \sum_{i=1}^k \text{Rep}_i. \quad (6.6)$$

Note also that the following equality holds

$$\int_0^\infty \ln |S(j\omega)| d\omega = \pi \sum_{i=1}^k \text{Rep}_i$$

if  $S(s)$  is a scalar function.

Theorem 6.2 has an important implication toward the limitations imposed by the open-loop unstable poles on sensitivity properties. It shows that there will exist a frequency range over which the largest singular value of the sensitivity function exceeds one if it is to be kept below one at other frequencies. In the presence of bandwidth constraint, this imposes a sensitivity tradeoff in different frequency ranges. Furthermore, this result suggests that unlike in single-input single-output systems, the limitations imposed by open-loop unstable poles in multivariable systems are related not only to the locations, but also to the directions of poles and their relative interaction. The case (i) of this result corresponds to the situation where the pole directions are mutually orthogonal, for which the integral corresponding to each singular value is related solely to one unstable pole with a corresponding distance to the imaginary axis, as if each channel of the system is decoupled from the others in effects of sensitivity properties. The case (ii) corresponds to the situation where all the pole directions are parallel, for which the unstable poles affect only the integral corresponding to the largest singular value, as if the channel corresponding to the largest singular value contains all unstable poles. Clearly, these phenomena are unique to multivariable systems. The following result further strengthens these observations and shows that the interaction between open-loop unstable poles plays an important role toward sensitivity properties.

**Corollary 6.3** *Let  $k = 2$  and let the Assumptions 6.1-6.2 hold. Then*

$$\int_0^\infty \ln \bar{\sigma}(S(j\omega)) d\omega \geq \frac{\pi}{2} \left( \text{Re}(p_1 + p_2) + \sqrt{(\text{Re}(p_1 - p_2))^2 + 4\text{Rep}_1\text{Rep}_2 \cos^2 \angle(\eta_1, \eta_2)} \right). \quad (6.7)$$

**Proof.** Note that

$$\begin{aligned} \lambda_{\max}((\text{Rep}_1)\eta_1\eta_1^* + (\text{Rep}_2)\eta_2\eta_2^*) &= \lambda_{\max} \left( \begin{bmatrix} \eta_1 & \eta_2 \end{bmatrix} \begin{bmatrix} \text{Rep}_1 & 0 \\ 0 & \text{Rep}_2 \end{bmatrix} \begin{bmatrix} \eta_1^* \\ \eta_2^* \end{bmatrix} \right) \\ &= \lambda_{\max} \left( \begin{bmatrix} \text{Rep}_1 & 0 \\ 0 & \text{Rep}_2 \end{bmatrix} \begin{bmatrix} \eta_1^* \\ \eta_2^* \end{bmatrix} \begin{bmatrix} \eta_1 & \eta_2 \end{bmatrix} \right) \\ &= \lambda_{\max} \left( \begin{bmatrix} \text{Rep}_1 & 0 \\ 0 & \text{Rep}_2 \end{bmatrix} \begin{bmatrix} 1 & \eta_1^*\eta_2 \\ \eta_2^*\eta_1 & 1 \end{bmatrix} \right) \\ &= \lambda_{\max} \left( \begin{bmatrix} \text{Rep}_1 & (\text{Rep}_1)\eta_1^*\eta_2 \\ (\text{Rep}_2)\eta_2^*\eta_1 & \text{Rep}_2 \end{bmatrix} \right). \end{aligned}$$

A straightforward calculation then gives

$$\lambda_{\max}((\text{Rep}_1)\eta_1\eta_1^* + (\text{Rep}_2)\eta_2\eta_2^*) = \frac{1}{2} \text{Re}(p_1 + p_2)$$

$$+ \frac{1}{2} \sqrt{(\operatorname{Re}(p_1 - p_2))^2 + 4\operatorname{Re}p_1\operatorname{Re}p_2 \cos^2 \angle(\eta_1, \eta_2)}.$$

The proof is now completed.  $\square$

The utility of this corollary is clear. This result fully characterizes the limitation imposed by a pair of open-loop unstable poles on the sensitivity reduction properties. This limitation depends not only on the relative distances of the poles to the imaginary axis, but also on the principal angle between the two pole directions.

Next, we investigate the design constraints imposed by open-loop non-minimum phase zeros upon sensitivity properties. The results below may be considered to be a matrix extension of the Poisson integral relation.

**Theorem 6.4** *Let  $S(s) \in \mathcal{H}_\infty$  be factorized as in (6.3) and assume that*

$$\lim_{R \rightarrow \infty} \max_{\phi \in [-\pi/2, \pi/2]} \frac{|\ln \overline{\sigma}(S(Re^{j\phi}))|}{R} = 0. \quad (6.8)$$

*Then, for any non-minimum phase zero  $z = x_0 + jy_0 \in \mathbb{C}_+$  of  $L(s)$  with output direction  $w$ ,  $w^*w = 1$ ,*

$$\int_{-\infty}^{\infty} \ln \overline{\sigma}(S(j\omega)) \frac{x_0}{x_0^2 + (\omega - y_0)^2} d\omega \geq \pi \ln \overline{\sigma}(S_m(z)) \geq \pi \ln \overline{\sigma}(S(z)). \quad (6.9)$$

$$\int_{-\infty}^{\infty} \ln \overline{\sigma}(S(j\omega)) \frac{x_0}{x_0^2 + (\omega - y_0)^2} d\omega \geq \pi \ln \|w^* B_k^{-1}(z) \cdots B_1^{-1}(z)\|. \quad (6.10)$$

Note that the condition (6.8) is satisfied if  $L(s)$  is a proper rational transfer matrix.

Furthermore, for single-input single-output systems ( $n = 1$ ),

$$\left\| w^* \prod_{i=1}^k B_i^{-1}(z) \right\| = \prod_{i=1}^k \left| \frac{z + \bar{p}_i}{z - p_i} \right|$$

and

$$\int_{-\infty}^{\infty} \ln |S(j\omega)| \frac{x_0}{x_0^2 + (\omega - y_0)^2} d\omega = \pi \ln \prod_{i=1}^k \left| \frac{z + \bar{p}_i}{z - p_i} \right|.$$

The more interesting result, however, is the inequality (6.10). This result again suggests that the multivariable sensitivity properties are closely related to the pole and zero directions. This result implies that the sensitivity reduction ability of the system may be severely limited by the open-loop unstable poles and non-minimum phase zeros, especially when these poles and zeros are close to each other and the angles between their directions are small.

### 6.3 Design Limitations and Sensitivity Bounds

The integral relations derived in the preceding section are now used to analyze the design tradeoffs and the limitations imposed by the bandwidth constraint and right-half plane poles and zeros upon sensitivity reduction properties. Similar to their counterparts for single-input single-output systems, these integral relations show that there will necessarily exist frequencies at which the sensitivity function exceeds one if it is to be kept below one over other frequencies, hence exhibiting a tradeoff between the reduction of the sensitivity over one frequency range against its increase over another frequency range.

Suppose that the feedback system is designed such that the level of sensitivity reduction is given by

$$\overline{\sigma}(S(j\omega)) \leq M_L < 1, \quad \forall \omega \in [-\omega_L, \omega_L], \quad (6.11)$$

where  $M_L > 0$  is a given constant. Let  $z = x_0 + jy_0 \in \mathbb{C}_+$  be an open right-half plane zero of  $L(s)$  with output direction  $w$ . Define also

$$\theta(z) := \int_{-\omega_L}^{\omega_L} \frac{x_0}{x_0^2 + (\omega - y_0)^2} d\omega.$$

The following lower bound on the maximum sensitivity displays a limitation due to the open right-half plane zeros upon the sensitivity reduction properties.

**Corollary 6.5** *Let the assumption in Theorem 6.4 holds. In addition, suppose that the condition (6.11) is satisfied. Then, for each open right-half plane zero  $z \in \mathbb{C}_+$  of  $L(s)$  with output direction  $w$ ,*

$$\|S(s)\|_\infty \geq \left( \frac{1}{M_L} \right)^{\frac{\theta(z)}{\pi - \theta(z)}} \|w^* B_k^{-1}(z) \cdots B_1^{-1}(z)\|^{\frac{\pi}{\pi - \theta(z)}}, \quad (6.12)$$

and

$$\|S(s)\|_\infty \geq \left( \frac{1}{M_L} \right)^{\frac{\theta(z)}{\pi - \theta(z)}} (\overline{\sigma}(S(z)))^{\frac{\pi}{\pi - \theta(z)}}. \quad (6.13)$$

**Proof.** Note that

$$\int_{-\infty}^{\infty} \ln \overline{\sigma}(S(j\omega)) \frac{x_0}{x_0^2 + (\omega - y_0)^2} d\omega \leq (\pi - \theta(z)) \ln \|S(j\omega)\|_\infty + \theta(z) \ln(M_L).$$

Then the inequality (6.12) follows by applying inequality (6.10) and inequality (6.13) follows by applying inequality (6.9)  $\square$

The interpretation of Corollary 6.5 is similar to that in single-input single-output systems. Roughly stated, this result shows that for a non-minimum phase system, its

sensitivity must increase beyond one at certain frequencies if the sensitivity reduction is to be achieved at other frequencies. Of particular importance here is that the sensitivity function will in general exhibit a larger peak in multivariable systems than in single-input single-output systems, due to the fact that  $\bar{\sigma}(S(z)) \geq 1$ .

The design tradeoffs and limitations on the sensitivity reduction which arise from bandwidth constraints as well as open-loop unstable poles can be studied using the extended Bode integral relations. However, these integral relations by themselves do not mandate a meaningful tradeoff between the sensitivity reduction and the sensitivity increase, since the sensitivity function can be allowed to exceed one by an arbitrarily small amount over an arbitrarily large frequency range so as not to violate the Bode integral relations. However, bandwidth constraints in feedback design typically require that the open-loop transfer function be small above a specified frequency, and that it roll off at a rate of more than one pole-zero excess above that frequency. These constraints are commonly needed to ensure stability robustness despite the presence of modeling uncertainty in the plant model, particularly at high frequencies. One way of quantifying such bandwidth constraints is by requiring the open-loop transfer function to satisfy

$$\bar{\sigma}(L(j\omega)) \leq \frac{M_H}{\omega^{1+k}} \leq \epsilon < 1, \quad \forall \omega \in [\omega_H, \infty) \quad (6.14)$$

where  $\omega_H > \omega_L$ , and  $M_H > 0$ ,  $k > 0$  are some given constants. With the bandwidth constraint given as such, the following result again shows that the sensitivity reduction specified by (6.11) can be achieved only at the expense of increasing the sensitivity at certain frequencies.

**Corollary 6.6** *Suppose the Assumptions 6.1-6.2 hold. In addition, suppose that the conditions (6.11) and (6.14) are satisfied for some  $\omega_H$  and  $\omega_L$  such that  $\omega_H > \omega_L$ . Then*

$$\max_{\omega \in [\omega_L, \omega_H]} \bar{\sigma}(S(j\omega)) \geq e^\alpha \left( \frac{1}{M_L} \right)^{\frac{\omega_L}{\omega_H - \omega_L}} (1 - \epsilon)^{\frac{\omega_H}{k(\omega_H - \omega_L)}} \quad (6.15)$$

where

$$\alpha = \frac{\pi \lambda_{\max} \left( \sum_{i=1}^k (\text{Rep}_i) \eta_i \eta_i^* \right)}{\omega_H - \omega_L}.$$

**Proof.** Note first that for  $\omega \geq \omega_H$ ,

$$\bar{\sigma}(S(j\omega)) = \frac{1}{\underline{\sigma}(I + L(j\omega))} \leq \frac{1}{1 - \bar{\sigma}(L(j\omega))} \leq \frac{1}{1 - \frac{M_H}{\omega^{1+k}}}$$

and

$$-\int_{\omega_H}^{\infty} \ln \left( 1 - \frac{M_H}{\omega^{1+k}} \right) d\omega = \sum_{i=1}^{\infty} \int_{\omega_H}^{\infty} \frac{1}{i} \left( \frac{M_H}{\omega^{1+k}} \right)^i d\omega$$



$$\begin{aligned}
&= \sum_{i=1}^{\infty} \frac{1}{i} \frac{\omega_H}{i(1+k)-1} \left( \frac{M_H}{\omega_H^{1+k}} \right)^i \\
&\leq \frac{\omega_H}{k} \sum_{i=1}^{\infty} \frac{1}{i} \left( \frac{M_H}{\omega_H^{1+k}} \right)^i = -\frac{\omega_H}{k} \ln \left( 1 - \frac{M_H}{\omega_H^{1+k}} \right) \\
&\leq -\frac{\omega_H}{k} \ln(1 - \epsilon).
\end{aligned}$$

Then

$$\begin{aligned}
\int_0^{\infty} \ln \bar{\sigma}(S(j\omega)) d\omega &= \int_0^{\omega_L} \ln \bar{\sigma}(S(j\omega)) d\omega + \int_{\omega_L}^{\omega_H} \ln \bar{\sigma}(S(j\omega)) d\omega + \int_{\omega_H}^{\infty} \ln \bar{\sigma}(S(j\omega)) d\omega \\
&\leq \omega_L \ln M_L + (\omega_H - \omega_L) \max_{\omega \in [\omega_L, \omega_H]} \ln \bar{\sigma}(S(j\omega)) - \int_{\omega_H}^{\infty} \ln \left( 1 - \frac{M_H}{\omega^{1+k}} \right) d\omega \\
&\leq \omega_L \ln M_L + (\omega_H - \omega_L) \max_{\omega \in [\omega_L, \omega_H]} \ln \bar{\sigma}(S(j\omega)) - \frac{\omega_H}{k} \ln(1 - \epsilon)
\end{aligned}$$

and the result follows from applying (6.4).  $\square$

The above lower bound shows that the sensitivity can be very significant in the transition band.

## 6.4 Bode's Gain and Phase Relation

In the classical feedback theory, the Bode's gain-phase integral relation (see Bode [1945]) has been used as an important tool to express design constraints in scalar systems. The following is an extended version of the Bode's gain and phase relationship for an open-loop stable scalar system with possible right-half plane zeros, see Freudenberg and Looze [1988] and Doyle, Francis, and Tannenbaum [1992]:

$$\angle L(j\omega_0) = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{d \ln |L|}{d\nu} \ln \coth \frac{|\nu|}{2} d\nu + \angle \prod_{i=1}^k \frac{j\omega_0 + \bar{z}_i}{j\omega_0 - z_i}$$

where  $z_i$ 's are assumed to be the right-half plane zeros of  $L(s)$  and  $\nu := \ln(\omega/\omega_0)$ . Note that  $\frac{d \ln |L(j\omega)|}{d\nu}$  is the slope of the Bode plot which is almost always negative. It follows that  $\angle L(j\omega_0)$  will be large if the gain  $L$  attenuates slowly and small if it attenuates rapidly. The behavior of  $\angle L(j\omega)$  is particularly important near the crossover frequency  $\omega_c$  where  $|L(j\omega_c)| = 1$  since  $\pi + \angle L(j\omega_c)$  is the phase margin of the feedback system, and further the return difference is given by

$$|1 + L(j\omega_c)| = |1 + L^{-1}(j\omega_c)| = 2 \left| \sin \frac{\pi + \angle L(j\omega_c)}{2} \right|$$

which must not be too small for good stability robustness. If  $\pi + \angle L(j\omega_c)$  is forced to be very small by rapid gain attenuation, the feedback system will amplify disturbances and exhibit little uncertainty tolerance at and near  $\omega_c$ . Since  $\angle \frac{j\omega_0 + \bar{z}_i}{j\omega_0 - z_i} \leq 0$  for each  $i$ , a non-minimum phase zero contributes an additional phase lag and imposes limitations upon the rolloff rate of the open-loop gain. The conflict between attenuation rate and loop quality near crossover is thus clearly evident. A thorough discussion of the limitations these relations impose upon feedback control design is given by Bode [1945], Horowitz [1963], and Freudenberg and Looze [1988]. See also Freudenberg and Looze [1988] for some multivariable generalizations.

In the classical feedback theory, it has been common to express design goals in terms of the “shape” of the open-loop transfer function. A typical design requires that the open-loop transfer function have a high gain at low frequencies and a low gain at high frequencies while the transition should be well-behaved. The same conclusion applies to multivariable system where the singular value plots should be well-behaved between the transition band.

## 6.5 Notes and References

The results presented in this chapter are based on Chen [1992a, 1992b, 1995]. Some related results can be found in Boyd and Desoer [1985] and Freudenberg and Looze [1988]. The related results for scalar systems can be found in Bode [1945], Horowitz [1963], Doyle, Francis, and Tannenbaum [1992], and Freudenberg and Looze [1988].

The study of analytic functions, harmonic functions<sup>1</sup>, and various integral relations in the scalar case can be found in Garnett [1981] and Hoffman [1962].

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<sup>1</sup>A function  $f : \mathbb{C} \rightarrow \mathbb{R}$  is said to be a harmonic function (subharmonic function) if  $\nabla^2 f(s) = 0$  ( $\nabla^2 f(s) \geq 0$ ) where the symbol  $\nabla^2 f(s)$  with  $s = x + jy$  denotes the Laplacian of  $f(s)$  and is defined by

$$\nabla^2 f(s) := \frac{\partial^2 f(s)}{\partial x^2} + \frac{\partial^2 f(s)}{\partial y^2}.$$